

TØ 8 – Mikroøkonomi 2 – 2024

Problems for Tutorial

1 Moral Hazard

Consider the following principal-agent model:

There is a risk-neutral principal with utility function $U^P = x - w$, where $x \in \{2, 7\}$ is the principal's revenue and w the wage she pays the agent. The principal makes a take-it-or-leave-it-offer to the agent. If the agent accepts the offer, he chooses an effort level $e \in \{e^l, e^h\}$. His utility is $U^A = \sqrt{w} - c(e)$, where $c(e) = \sqrt{e}$. Note that $e^h > e^l \geq 0$. The effort level stochastically determines the principal's revenue:

$$Pr(x = 7|e^h) = \frac{3}{4} \text{ and } Pr(x = 7|e^l) = \frac{1}{2}.$$

If the agent rejects the offer, he receives his reservation utility $\underline{u} = 0$.

Part A: For questions (1)-(4) suppose the agent's effort is observable and verifiable.

1. Which variables can wages condition on?
2. State the principal's problem and explain its different elements.
3. Derive the FOCs and reflect on the optimal wage scheme for each possible effort level the principal may implement. Be sure to specify the full wage contract conditioning on any variables from your answer to (1).
4. Solve the last part of the principal's problem, i.e., determine which contract the principal offers. Interpret your findings (3-4 sentences).

Note til moral hazard (MH): min måde at gøre det på er en smule anderledes end Marie-Louises, fordi jeg har lavet meget MH i det fag, der hedder Micro 2 (som er en stor banger i øvrigt. Kan klart anbefales), og her var fremgangsmåden en lidt anden. Man kommer frem til de samme resultater, og jeg forstår udmærket jeres metoder, hvis I løser som Marie-Louise, men hvis I er i tvivl om noget vedrørende mine løsninger, skal I bare skrive eller sige til.

1

Wages under first best can depend on the level effort and the outcome.

2

Let $p_i(e)$ be the probability of a certain outcome given a certain effort, e.g., $p_h(e^h)$ is the probability of reaching the high level of output conditional on the agent providing high effort. The principal's optimisation problem if she wants to implement **high effort** is

$$\begin{aligned} \max p_h(e^h) (x_H - w_H) + (1 - p_h(e^h))(x_L - w_L) \\ \text{s. t. } p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_L) - c(e_h) \geq \bar{u} \end{aligned}$$

Where I have defined $u(w) \equiv \sqrt{w}$

The maximisation problem can be rewritten into (and this applies to every MH problem):

$$\begin{aligned} \max p_h(e^h) (x_H - w_H) + (1 - p_h(e^h))(x_L - w_L) \\ = \max p_h(e^h) \cdot (-w_H) + (1 - p_h(e^h)) \cdot (-w_L) \\ = \min p_h(e^h) \cdot w_H + (1 - p_h(e^h)) \cdot w_L \end{aligned}$$

Since $p_h(e^h) \cdot x_i$ is a constant, and constants can be omitted when optimising. Wages, on the other hand, are of course not constants

And likewise, if the principal wants to implement **low effort**:

$$\begin{aligned} \min p_h(e^l) \cdot w_H + (1 - p_h(e^l)) \cdot w_L \\ \text{s. t. } p_h(e^l)u(w_H) + (1 - p_h(e^l))u(w_L) - c(e_l) \geq \bar{u} \end{aligned}$$

Explanation of the problem: principal wants to maximise profits subject to the PC, i.e., that the agent will work rather than declining the job offer

3

Lagrangian when high effort is demanded is

$$\begin{aligned} \mathcal{L} = p_h(e^h) \cdot w_H + (1 - p_h(e^h)) \cdot w_L \\ - \lambda [p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_L) - c(e) - \bar{u}] \end{aligned}$$

FOCs

$$\begin{aligned} w_H: p_h(e^h) - \lambda p_h(e^h)u'(w_H) = 0 &\Leftrightarrow u'(w_H) = \frac{1}{\lambda} \\ w_L: p_h(e^h) - \lambda p_h(e^h)u'(w_L) = 0 &\Leftrightarrow u'(w_L) = \frac{1}{\lambda} \end{aligned}$$

Which implies that $u'(w_H) = u'(w_L) \Rightarrow u(w_H) = u(w_L) \Rightarrow w_H = w_L = w$

Similar if low effort is demanded

$$\mathcal{L} = p_h(e^l) \cdot w_H + (1 - p_h(e^l)) \cdot w_L - \lambda [p_h(e^l)u(w_H) + (1 - p_h(e^l))u(w_L) - c(e) - \bar{u}]$$

FOCs

$$w_H: p_h(e^h) = \lambda p_h(e^h)u'(w_H) \Leftrightarrow u'(w_H) = \frac{1}{\lambda}$$

$$w_L: p_h(e^h) = \lambda p_h(e^l)u'(w_L) \Leftrightarrow u'(w_L) = \frac{1}{\lambda}$$

The conclusion is that the wage will be the same if the profit is high or low, both if low or high effort is demanded—this is also called full insurance

4

The problem is solved by inserting w into the binding PC

$$p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_L) - c(e_h) = \bar{u} \Rightarrow \frac{3}{4} \cdot \sqrt{w} + \frac{1}{4} \cdot \sqrt{w} - \sqrt{e_h} = 0 \Leftrightarrow w = e_h$$

And if low effort is demanded

$$p_h(e^l)u(w_H) + (1 - p_h(e^l))u(w_L) - c(e) = \bar{u} \Rightarrow \frac{1}{2} \sqrt{w} + \frac{1}{2} \sqrt{w} - \sqrt{e_l} = 0 \Leftrightarrow w = e_l$$

And the optimal wage scheme is therefore:

When high effort is demanded

$$w(e^h) = \begin{cases} e^h, & e = e^h \\ e^h - f, & e = e^l \end{cases}$$

And when low effort is demanded

$$w(e^l) = \begin{cases} e^l, & e = e^l \\ e^l - f, & e = e^h \end{cases}$$

Where f is a fine given to the agent if he does not provide the effort demanded by the principal. The fine must satisfy

$$u(w) - c(e) \geq u(w - f) - c(e')$$

Where e' is the other level of effort, i.e., the one the principal does not want implemented

In words, the utility from the wage received minus the cost of demanded effort must be greater than or equal to the utility from receiving the wage but paying a fine minus the other cost of effort

Comparing the profits under high and low effort (with $w_H = w_L = w$) suggests that the optimal contract demands high effort if and only if

$$p_h(e^h)(x_H - w) + (1 - p_h(e^h))(x_L - w) > p_h(e^l)(x_H - w) + (1 - p_h(e^l))(x_L - w)$$

Simplifying a bit and inserting the optimal wage found before

$$p_h(e^h)x_H + (1 - p_h(e^h))x_L - e_h > p_h(e^l)x_H + (1 - p_h(e^l))x_L - e_l$$

Inserting the numbers:

$$\begin{aligned} \frac{3}{4} \cdot 7 + \frac{1}{4} \cdot 2 - e_h &> \frac{1}{2} \cdot 7 + \frac{1}{2} \cdot 2 - e_l \Leftrightarrow \frac{21}{4} + \frac{2}{4} - e_h > \frac{7}{2} + 1 - e_l \Leftrightarrow \frac{23}{4} - e_h > \frac{9}{2} - e_l \\ &\Leftrightarrow \frac{5}{4} > e_h - e_l \end{aligned}$$

So if $\frac{5}{4} > e_h - e_l$, high effort is optimal to implement

Part B: For question (5)-(6) suppose the agent's effort is unobservable and non-verifiable. Further assume that $e^h = 1$ and $e^l = \frac{4}{9}$.

5. Which variables can wages condition on? State the principal's problem, and explain each element (write at most 2 sentences per element).
6. Would the contract derived in Part A still work as intended? Explain why/why not (3-4 sentences).

5

Because the principal cannot verify the agent's effort, i.e., principal cannot observe e_H or e_L , the wage will now depend on the outcome (profit) rather than the effort.

If the principal wants to implement high effort, the minimisation problem is

$$\begin{aligned} \min & p_h(e^h) \cdot w_H + (1 - p_h(e^h)) \cdot w_L \\ \text{s.t.} & p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_L) - c(e_h) \geq \bar{u} \text{ (PC)} \\ & p_h(e^h)u(w_H) + (1 - p_h(e^h))u(w_L) - c(e_h) \\ & \geq p_h(e^l)u(w_H) + (1 - p_h(e^l))u(w_L) - c(e_l) \geq \bar{u} \text{ (IC)} \end{aligned}$$

The IC can be rewritten into:

$$\left(p_h(e^h) - p_h(e^l)\right)u(w_H) - \left(\left(1 - p_h(e^h)\right) - \left(1 - p_h(e^l)\right)\right)u(w_L) - \left(c(e^h) - c(e^l)\right) \geq 0$$

Where the second constraint is the incentive compatibility constraint (IC). Since effort is unobservable, it must be optimal for the agent to provide high effort rather than deviating by providing low effort, if high effort is demanded by the principle.

6

This can be proven using the IC and the fact that under FB, we had the contract: $w_L = w_H = w$:

$$\begin{aligned} \left(\frac{3}{4} - \frac{1}{2}\right)u(w_H) - \left(\frac{1}{4} - \frac{1}{2}\right)u(w_L) - \left(1 - \frac{2}{3}\right) &= \left(\frac{3}{4} - \frac{1}{2}\right)u(w) - \left(\frac{1}{4} - \frac{1}{2}\right)u(w) - \left(1 - \frac{2}{3}\right) \\ &= \frac{1}{4}u(w) - \frac{1}{4}u(w) - 1 + \frac{2}{3} \geq 0 \Rightarrow -\frac{1}{3} \geq 0 \end{aligned}$$

Which is not satisfied

Insurance

A risk averse individual (agent A) with initial wealth w has a new car valued L subject to a risk of being stolen. In order to protect himself in the case of the car being stolen, the individual can buy insurance from a risk neutral insurance company (principal P). The company and the individual share the same common prior of the risk of the car being stolen. In particular the car is stolen with probability k , which we here assume constant and independent of any actions of the agent.

The individual chooses coverage q such that he either insures fully ($q = L$) or only partly ($q < L$). The insurance company demands a price (or a premium) p for providing insurance coverage q . The price is paid regardless of whether the car is stolen or not.

The car is valued as an asset to the individual and thus affects his utility through the impact on his total wealth, i.e. in the absence of any insurance the agent gets utility $u(w)$ when the car is stolen and $u(w + L)$ when the car is not stolen.

1. Determine A's *reservation utility* (outside option). I.e. the utility A gets from not participating in the insurance market.
2. What could P include as verifiable terms in the insurance contract? (what does the contract depend on)
3. Assume P decides on p and q . State the optimization problem and derive FOC.
4. Solve for the q and p which the principal chooses in optimum. Do P earn positive profits? Is the outcome efficient?

1

Reservation utility is

$$\bar{u} = k \cdot u(w) + (1 - k) \cdot u(w + L)$$

That is, the probability k of the car being stolen times his utility. In other words: he gains L in utility from the car not being stolen

2

The price (or premium) offered by the insurance company is verifiable. So is the coverage chosen by the agent. Factors such as initial wealth, car's value, and the probability of it being stolen are of course not directly observable to the insurance company.

3

Maximization problem of the principal is

$$\max(1 - k) \cdot p + k(p - q) = p - kq$$

$$s. t. k \cdot u(w - p + q) + (1 - k)u(w + L - p) \geq \bar{u}$$

That is, the probability of the car not being stolen, and the car being stolen must be chosen such that it is equal to the price paid by the customer minus the probability of it being stolen times the coverage the insurance company must pay if it happens.

Subject to the PC, i.e., the agent must get a higher utility by accepting the insurance offer than if he did not ($\bar{u}$ is defined in 1).

Lagrange:

$$\mathcal{L} = p - kq + \lambda((1 - k)u(w + L - p) + k \cdot u(w - p + q) - \bar{u})$$

FOCs are

$$\frac{\partial \mathcal{L}}{\partial p} = 1 - \lambda((1 - k)u'(w + L - p) + ku'(w - p + q)) = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} = -k + \lambda(ku'(w - p - q)) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = (1 - k)u(w + L - p) + k \cdot u(w - p + q) = 0$$

4

FOCs can be rewritten into

The second one:

$$-k + \lambda(ku'(w - p - q)) = 0 \Leftrightarrow \lambda = \frac{1}{u'(w - p - q)}$$

So, we can substitute the value for λ into the first one

$$\begin{aligned} 1 - \lambda((1 - k)u'(w + L - p) + ku'(w - p + q)) \\ = 1 - \frac{1}{u'(w - p - q)}((1 - k)u'(w + L - p) + ku'(w - p + q)) = 0 \Leftrightarrow \\ 1 = \frac{1}{u'(w - p - q)}((1 - k)u'(w + L - p) + ku'(w - p + q)) \end{aligned}$$

And now, since (in the first equation), we can insert the value of lambda

$$\begin{aligned} 1 - \lambda((1 - k)u'(w + L - p) + ku'(w - p + q)) \\ = 1 - \frac{1}{u'(w - p - q)}((1 - k)u'(w + L - p) + ku'(w - p + q)) = 0 \end{aligned}$$

$$\Leftrightarrow \frac{1}{u'(w-p-q)}((1-k)u'(w+L-p) + ku'(w-p+q)) = 1$$

$$\Leftrightarrow u'(w-p-q) = (1-k)u'(w+L-p) + ku'(w-p+q)$$

But the above equation implies that

$$u'(w-p-q) = u'(w+L-p)$$

So therefore, $q = L$

This can be used in the PC (binding since in optimum)

$$(1-k)u(w+L-p) + k \cdot u(w-p+q) = (1-k) \cdot u(w+L-p) + k \cdot u(w-p+L)$$

$$= u(w+L-p)$$

$$= \bar{u}$$

$$\Leftrightarrow w+L-p = u^{-1} \cdot \bar{u}$$

$$\Leftrightarrow -w-L+p = -u^{-1} \cdot \bar{u}$$

$$p = w+L - u^{-1} \cdot \bar{u}$$

Profits are then

$$p - kq = w + L - u^{-1}(\bar{u}) - k \cdot L = w + (1-k)L - u^{-1}(\bar{u}) > 0$$

Since $p = w + L - u^{-1} \cdot \bar{u} > 0$ by definition, and therefore $p - kq = w + (1-k)L - u^{-1}(\bar{u}) > 0$ as we do not have perfect competition (just yet)

It is efficient since the agent receives full coverage and the principal's profits are positive

5. Assume P competes in a perfectly competitive market with other insurance companies. What price p would P charge?
6. What coverage q would A prefer with the price being set as in 5)? Is the outcome efficient?
7. What contract does P offer if A is risk neutral ($u(w) = w$)?
8. Assume that both P and A are risk averse (P 's utility function is $W(\pi)$). Would P offer full coverage $q = L$?

5

In a perfect market, the profit will be zero (just basic micro 1) and hence,

$$\max(1-k) \cdot p + k(p-q) = p - kq = 0 \Rightarrow p = kq$$

6

Since the agent is risk averse, he will choose full coverage, so the maximisation of the agent is

$$\max ku(w + q - kq) + (1 - k)u(w + L - kq)$$

Lagrange (not really necessary, just find the derivative wrt q)

$$\mathcal{L} = ku(w + q - kq) + (1 - k)u(w + L - kq)$$

FOC is

$$\frac{\partial \mathcal{L}}{\partial q} = -(1 - k)u'(w + L - kq)k + (1 - k)ku'(w - kq + q) = 0$$

Which can be simplified:

$$\begin{aligned} & -(1 - k)u'(w + L - kq)k + (1 - k)ku'(w - kq + q) \\ & = (-1 + k)u'(w + L - kq)k + (1 - k)ku'(w - kq + q) \\ & = u'(w + L - kq)k + ku'(w - kq + q) \\ & = u'(w + L - kq) + ku'(w - kq + q) = 0 \\ & \Leftrightarrow u'(w + L - kq) = u'(w - kq + q) \Rightarrow L = q \end{aligned}$$

That is, no matter the probabilities, he will always choose full coverage. If it is stolen, he pays price kq but receives q in premium. If it is not stolen, he gains the L (his valuation) but still pays price kq and, needless to say, does not receive the premium.

It is efficient since q must be chosen such that $q \leq L$ which is satisfied

7

If the agent is risk-neutral, $u(w) = w$, we can simplify the reservation utility into

$$\bar{u} = k \cdot u(w) + (1 - k) \cdot u(w + L) = kw + (1 - k)(w + L) = w + (1 - k)L$$

Utility from accepting the offer is

$$\begin{aligned} (1 - k) \cdot u(w + L - p) + k \cdot u(w - p + q) &= (1 - k)(w + L - p) + k \cdot (w - p + q) \\ &= w - p + (1 - k) \cdot L + k \cdot q \end{aligned}$$

And we insert this into the PC:

$$w - p + (1 - k) \cdot L + k \cdot q \geq \bar{u} = w + (1 - k)L \Leftrightarrow kq \geq p$$

What does this mean?

When $kq > p$, negative profits

When $kq = p$, the firm earns zero profit regardless of the level of coverage. Thus, both the firm and the agent are indifferent between each level of coverage.

8

π and p are equivalent. Very confusing I know. I will suggest that the description should be changed

$$\begin{aligned} \max & (1-k)W(p) + kW(p-q) \\ \text{s.t.} & (1-k)u(w+L-p) + ku(w-p+q) \geq \bar{u} \end{aligned}$$

Lagrangian

$$\mathcal{L} = (1-k)W(p) + kW(p-q) + \lambda[(1-k)u(w+L-p) + ku(w-p+q) - \bar{u}]$$

FOCs

$$\frac{\partial \mathcal{L}}{\partial p} = (1-k)W'(p) + kW'(p-q) - \lambda[(1-k)u'(w+L-p) + ku'(w-p+q)] = 0$$

$$\frac{\partial \mathcal{L}}{\partial q} = -kW'(p-q) + \lambda[ku'(w-p+q)] = 0$$

Rewrite the FOCs

$$(1-k)W'(p) + kW'(p-q) = \lambda[(1-k)u'(w+L-p) + ku'(w-p+q)]$$

$$kW'(p-q) = \lambda[ku'(w-p+q)]$$

Combine the two:

$$\frac{(1-k)W'(p) + kW'(p-q)}{kW'(p-q)} = \frac{\lambda[(1-k)u'(w+L-p) + ku'(w-p+q)]}{\lambda[ku'(w-p+q)]}$$

Simplify the LHS

$$\frac{(1-k)W'(p) + kW'(p-q)}{kW'(p-q)} = \frac{(1-k)W'(p)}{kW'(p-q)} + \frac{kW'(p-q)}{kW'(p-q)} = \frac{(1-k)W'(p)}{kW'(p-q)} + 1$$

And RHS

$$\begin{aligned}\frac{\lambda[(1-k)u'(w+L-p) + ku'(w-p+q)]}{\lambda[ku'(w-p+q)]} &= \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)} + \frac{ku'(w-p+q)}{ku'(w-p+q)} \\ &= \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)} + 1\end{aligned}$$

Putting these together yields

$$\frac{(1-k)W'(p)}{kW'(p-q)} + 1 = \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)} + 1$$

$$\Leftrightarrow \frac{(1-k)W'(p)}{kW'(p-q)} = \frac{(1-k)u'(w+L-p)}{ku'(w-p+q)}$$

$$\Leftrightarrow \frac{W'(p)}{W'(p-q)} = \frac{u'(w+L-p)}{u'(w-p+q)}$$

The last line: multiply by $\frac{k}{1-k}$ on both sides of the equality

Would P then offer full coverage, i.e., $q = L$?

$$\begin{aligned}\frac{W'(p)}{W'(p-q)} = \frac{u'(w+L-p)}{u'(w-p+q)} &\Rightarrow \frac{W'(p)}{W'(p-L)} = \frac{u'(w+L-p)}{u'(w-p+L)} = 1 \Leftrightarrow W'(p) = W'(p-L) \\ &\Leftrightarrow L = 0\end{aligned}$$

But $L > 0$, so P will not offer $q = L$

9. Assume $\frac{k}{1-k} = 1$, $U(w) = \ln(w)$, $W(\pi) = \ln(\pi)$, $W=10$ and $L=5$. What coverage would P offer?

Use the modified FOC to obtain:

$$\frac{W'(p)}{W'(p-q)} = \frac{u'(w+L-p)}{u'(w-p+q)} \Leftrightarrow \frac{p}{p-q} = \frac{w+L-p}{w-p+q} \Leftrightarrow p = q \left(\frac{w+L}{L} \right) = q \left(\frac{10+5}{5} \right) = 3q$$

We use the binding PC (because PC binds in optimum) to solve for p

LHS of the PC is

$$(1-k)u(w+L-p) + ku(w-p+q) = (1-k)\ln(w+L-p) + k\ln(w-p+q)$$

And RHS, the reservation utility:

$$\bar{u} = k \cdot u(w) + (1 - k) \cdot u(w + L) = k \cdot \ln(w) + (1 - k) \cdot \ln(w + L)$$

Putting the two together

$$\begin{aligned} (1 - k) \ln(w + L - p) + k \ln(w - p + q) &= k \ln(w) + (1 - k) \ln(w + L) \\ \Leftrightarrow (1 - k) \ln(w + L - 3q) + k \ln(w - 3q + q) &= k \ln(w) + (1 - k) \ln(w + L) \\ \Leftrightarrow (1 - k) \ln(15 - 3q) + k \ln(10 - 2q) &= k \ln(10) + (1 - k) \ln(15) \\ \Leftrightarrow \ln(15 - 3q) + \ln(10 - 2q) &= \ln(10) + \ln(15) \Leftrightarrow q \in \{0, 10\} \end{aligned}$$

Last step: because $\frac{k}{1-k} = 1 \Rightarrow k = 1 - k$

So, $q = 0$ or $q = 10$

Notice that the second solution yields $\ln(w + L - p) = \ln(w + L - 3q) = \ln(10 + 5 - 3 \cdot 10) = \ln(-15)$ which is undefined. Hence, $q = 0$ is the only valid solution

Since both parties dislike risk equally—both have the utility function $\ln(\cdot)$ —the principal will charge a very high price of insurance, so high that profit, π or p , will never be close to 0. In other words, the price of the coverage will—because of a risk-averse principal—be so high that even a risk-averse agent will find it ridiculous. The agent, even though being risk-averse, will therefore not agree to the contract

3 Out-of-court settlement

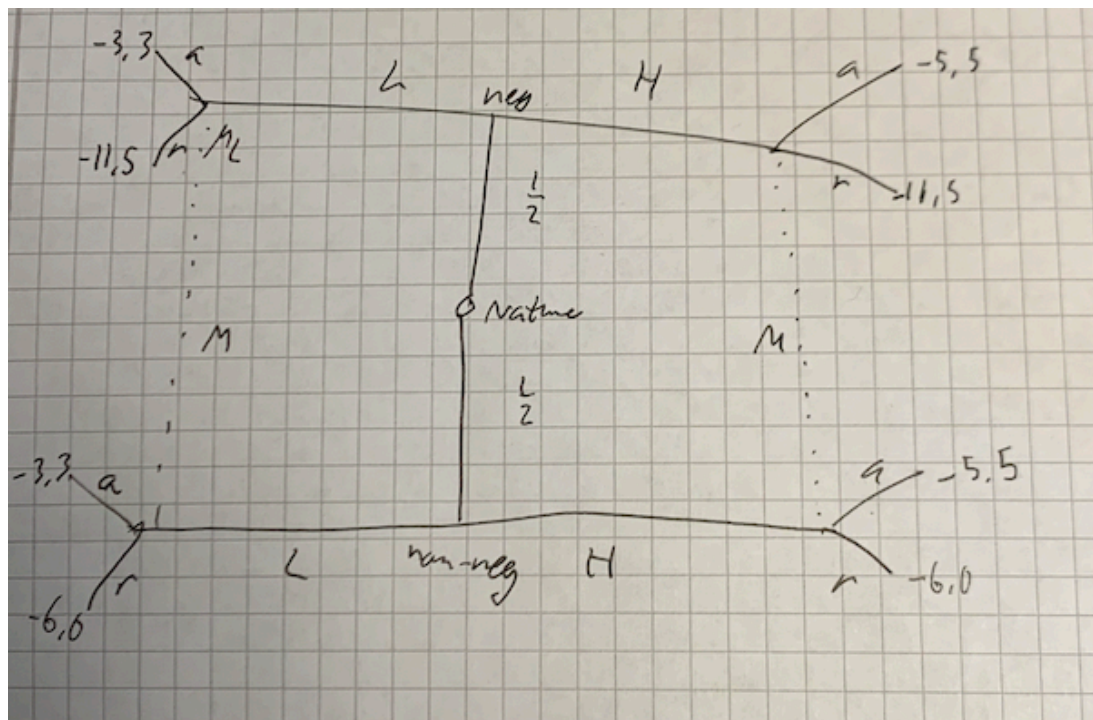
(25 points) A dam that a construction firm built for a municipality has collapsed. The construction firm knows whether the collapse was due to negligence on its part or due to circumstances outside of its control. The municipality does *not* observe which of these two states is the true one. Both sides know that in a court trial, the true state will eventually be uncovered.

After observing the state of the world, the firm tries to arrive at a out-of-court settlement with the municipality. It has two alternatives: to make a low settlement offer, under which it pays \$300,000, or a high offer, under which it pays the municipality \$500,000.

After the firm has submitted the settlement offer, the municipality decides whether to accept it or not. In case that the municipality rejects the offer, the matter is settled in a court trial. Both parties are aware of the fact that the firm's lawyer fees in case of a trial will be \$600,000, and that, in addition to this, if the firm is found guilty of negligence, it will be required to pay the municipality \$500,000 in damages. The municipality has no lawyer costs to pay in case of a trial.

1. Draw the game tree that describes this situation, assuming that the root of the game tree is a move by nature that assigns equal probability to the firm being negligent or non-negligent.
2. Find all the pure-strategy Perfect Bayesian equilibria (PBE) of the game specified in (1).
3. Which of the above equilibria satisfy the intuitive criterion? Please explain (write max. 6 sentences).

1



2

I start with the pooling

LL:

$$EU_2[a|\mu_L] = \frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 3 = 3$$

$$EU_2[r|\mu_L] = \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 0 = \frac{5}{2}$$

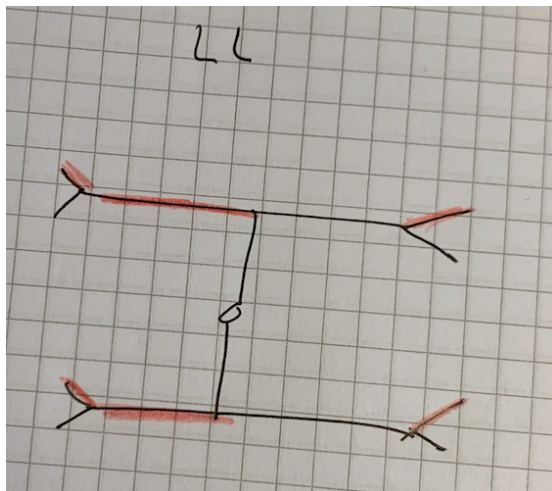
So, $EU_2[a|\mu_L] > EU_2[r|\mu_L]$

And in the RHS of the game, $\mu_H \in [0,1]$ since it is a weakly dominated strategy for player 2 to choose a , so we do not need to restrict μ_H

This is a PBE since if player 1 deviated, she would get a payoff of $-5 < -3$

The PBE is

$$PBE = \left(LL, aa, \mu_L = \frac{1}{2}, \mu_H \in [0,1] \right)$$



HH:

$$EU_2[a|\mu_H] = \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 5 = 5$$

$$EU_2[r|\mu_H] = \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 0 = \frac{5}{2}$$

Meaning that once again, $EU_2[a|\mu_H] > EU_2[r|\mu_H]$, so player 2 will choose a when player 1 chooses H

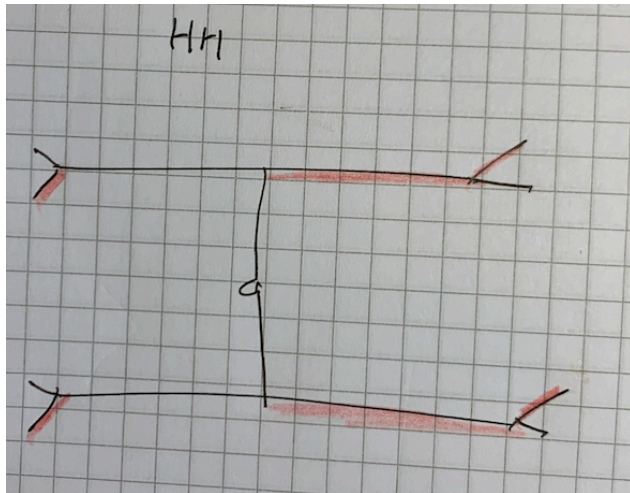
We need to restrict μ_L since there could be a possible deviation in the LHS of the figure: if player 2 chooses a here, player 1 could deviate and get a payoff of $-3 > -5$

So, μ_L needs to satisfy the following

$$5\mu_L + 0(1 - \mu_L) \geq 3\mu_L + 3(1 - \mu_L) \Leftrightarrow \mu_L \geq \frac{3}{5}$$

And hence, the PBE is

$$PBE = \left(HH, ra, \mu_L \geq \frac{3}{5}, \mu_H = \frac{1}{2} \right)$$



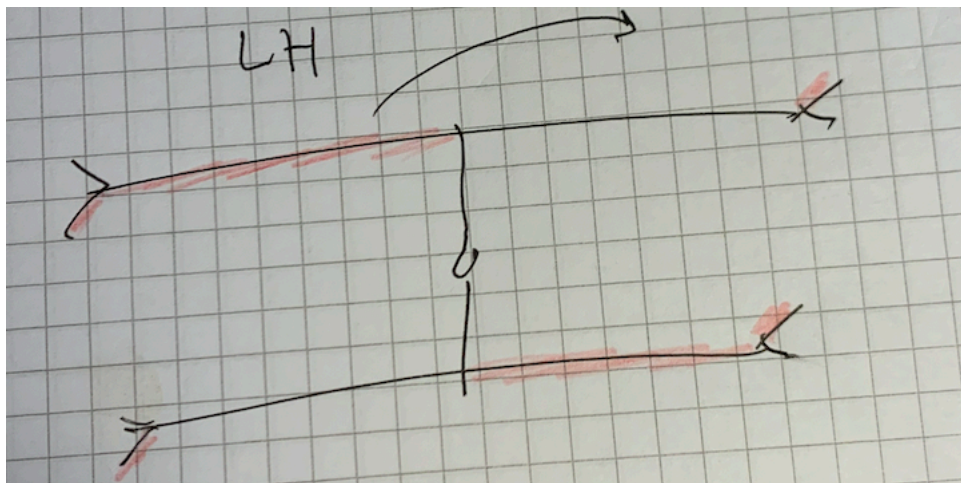
LH:

In L, player 2 will choose r since $5 > 3$

In H, player 2 will choose a as well: $5 > 0$

In the neg scenario (upper part of figure) player 1 will deviate since player 1 gets -11 in payoff and -5 if she deviates (shown by the arrow)

Therefore, there does not exist a PBE where the neg type chooses L and the non-neg type chooses H



HL:

In the upper part, player 2 chooses either a or r: she is indifferent

Which is not satisfied either. In conclusion, the PBE LL survives the intuitive criterion

How about HH?

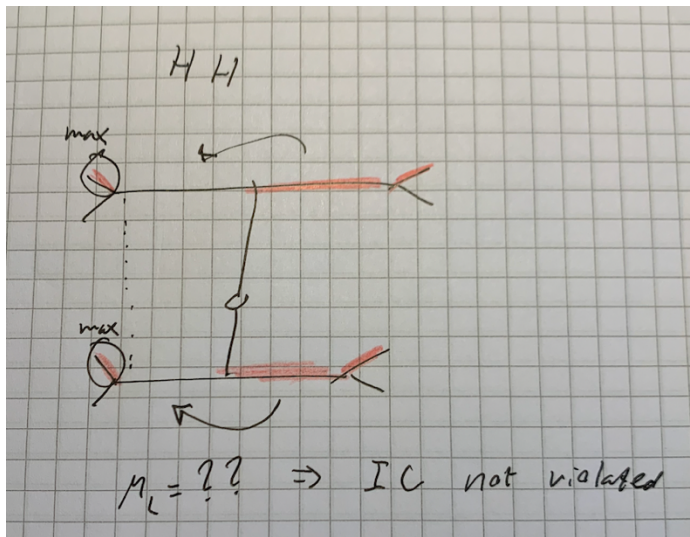
In the top part:

$$\max u_1(L|neg) > u_1(H|neg) \Leftrightarrow -3 > -5$$

And bottom part

$$\max u_1(L|non - neg) > u_1(H|non - neg) \Leftrightarrow -3 > -5$$

So, since the maximum payoff of both types is greater than the payoff from the equilibrium path, the μ_L cannot be restricted. In other words, we cannot say the deviating choice L is 100% from the neg type or 100% from the non-neg type and hence, $\mu_L \neq 1, 1 - \mu_L \neq 1$. Therefore, it survives the intuitive criterion.



If, say, the payoff in neg, L (upper left part) was for player 1 -8 instead of -3 , the inequality would be

$$\max u_1(L|neg) > u_1(H|neg) \Leftrightarrow -8 > -5$$

Which is not satisfied, marked with a cross on the drawing

This would imply that the belief μ_L could be restricted, as only one of the types of player 1 (the non-neg type player 1) could possibly deviate, i.e., $\mu_L = 0$

And in non-neg L, player 2 would choose a since his payoff is greater than choosing r: $3 > 0$

So, in conclusion, this modification would violate the intuitive criterion

